

Workshop on Computational Fluid Dynamics (CFD)
with OpenFOAM
IMP-PAN Gdańsk

Tutorial #2: Water-air Ejector
Multiphase Transient Flow

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Learning Objectives

By the end of this tutorial the student will be able to

1. **Navigate** OpenFOAM case directory structure
2. **Understand** basic OpenFOAM file formats
3. **Run** a simple transient simulation
4. **Analyze** basic results in terminal and **paraview**

1 Introduction and Theoretical Background

1.1 Case Overview

2 Case Setup

2.1 Python virtual environment

To facilitate the generation of the mesh with **blockMesh** we are going to use a python code. It is convenient to create first a user virtual environment, so that packages can be locally installed. Here follows the procedure

```
# Create the virtual environment. Here the name is 'my_env' but it can be any
name
python -m venv my_env
# Activate the virtual environment
source ./my_env/bin/activate
# From now on a '(my_env)' will be shown with prompt
# To deactivate the virtual environment, just type
deactivate
```

The module [ofblockmeshdicthelper](#), by A. Takaaki will be used. It has to be installed in the environment with

```
pip install 'ofblockmeshdicthelper @git+https://github.com/takaakiaoki/
ofblockmeshdicthelper.git'
```

If some other module is needed, just install it with **pip install <module name>**.

3 Part A: Volume of Fluid Approach

The clean case can be downloaded from the GitHub repository,

3.1 Running the Simulation

The simulation process is split into two automated phases: mesh generation and solver execution. Both phases are implemented in shell scripts, to ensure consistent and reproducible workflow.

3.1.1 Mesh generation

The first step is to create the `system/blockMeshDict` file. It is done with the python code `blockMeshDictGen.py`, that load the `ofblockmeshdicthelper` mentioned in the previous section.

Check that the python virtual environment is activated and run

```
python blockMeshDictGen.py
```

The file `system/blockMeshDict` has been created.

Run the script for the mesh generation

```
./genMesh.scr
```

WARNING: If the command fails, it is probable that it is not executable. Change it with

```
chmod +x ./genMesh.scr
```

Have a look at the script and discuss the meaning of each command. Check the output of the `checkMesh` command and visualize the mesh with `paraFoam`

3.1.2 Initialization and solver execution

The `runCase.scr` performs several tasks:

1. It copies the content of the folder `0.orig` to the folder `0`.
2. It initializes the phase distribution with the command `setFields` and the dictionary `system/setFieldsDict`
3. It runs the `interFoam` solver.

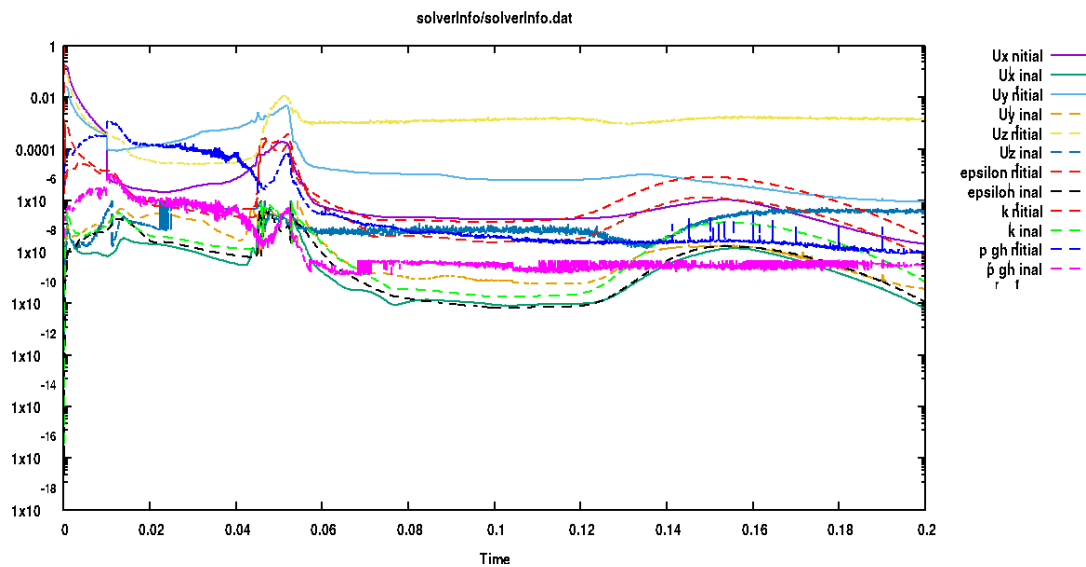
The simulation will take about 10 minutes. In the meanwhile the configuration files of the case can be studied:

- `system/controlDict` and functions definitions
- `0.orig/*`
- `constant/*`
- `system/fvSolution` and `system/fvSchemes`

3.2 Post-Processing in Terminal

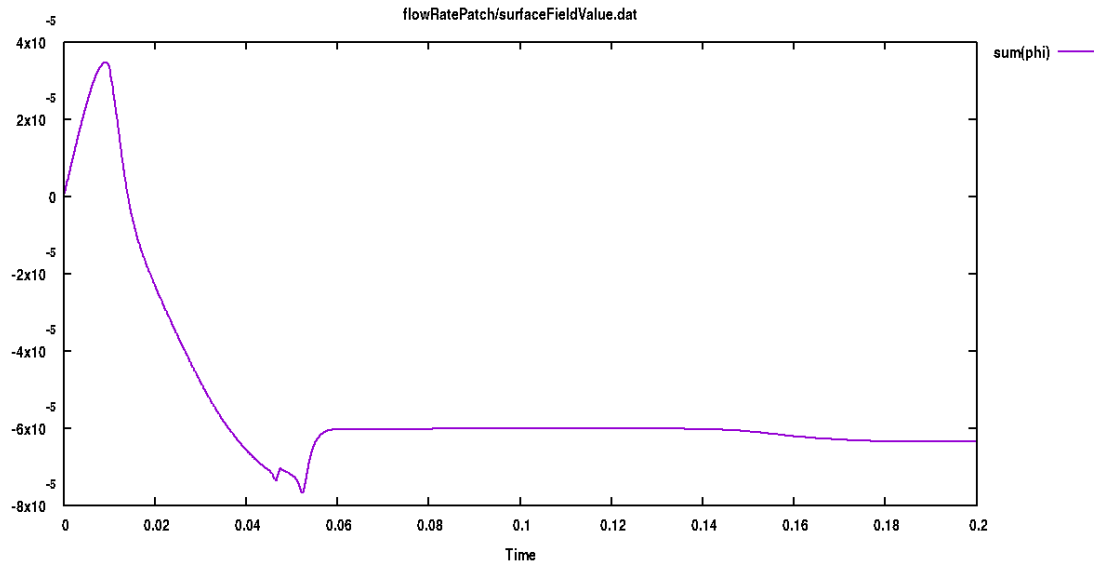
Residuals of the simulation can be check with

```
foamMonitor -l postProcessing/solverInfo/0/solverInfo.dat
```

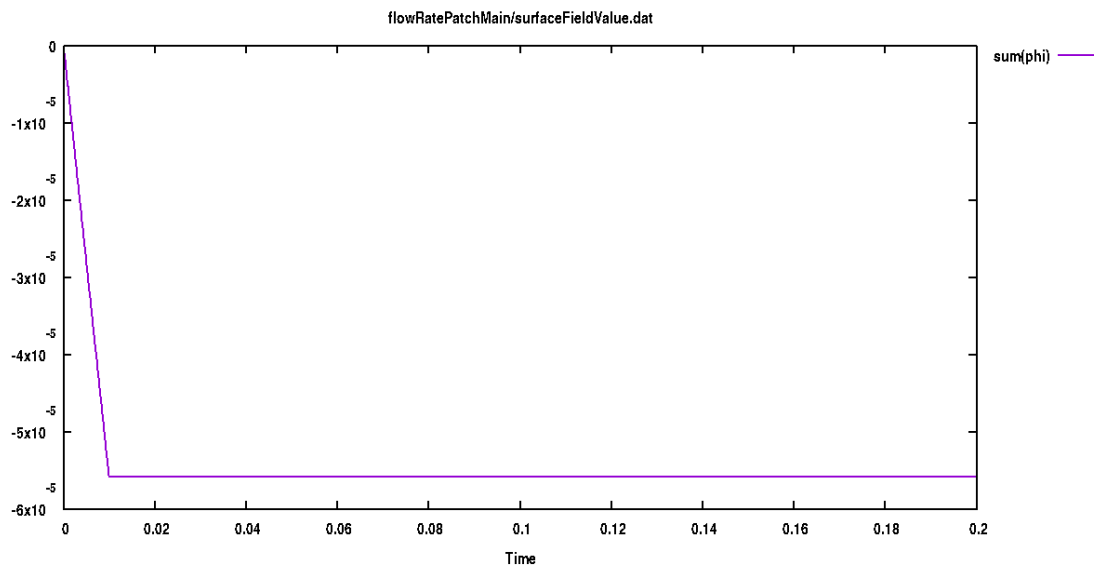


Also flow rates from main and secondary inlets

```
foamMonitor postProcessing/flowRatePatch/0/surfaceFieldValue.dat
```



```
foamMonitor postProcessing/flowRatePatchMain/0/surfaceFieldValue.dat
```



Note that the flow rate in main secondary inlets are¹

$$Q_s = 6.35 \times 10^{-5} \text{ m}^3/\text{s}$$

$$Q_m = 5.58 \times 10^{-5} \text{ m}^3/\text{s}$$

¹These are the flow rate for **wedge** of the mesh. For the complete flow rate, these numbers have to be multiplied by the proportionality factor of total and wedge sections.

According to that the dimensionless **entrainment ratio** is

$$M = \frac{Q_s}{Q_m} = 1.14$$

Pressure in the inlet patch can also be visualized (do it by yourself)

3.3 Visualization in paraview

Progress of water and air entrainment in the ejector can be seen in paraview, and even an animation can be created with the **foamCreateVideo** utility. Phase fraction and velocity distributions are shown in Figures 1 and 2.

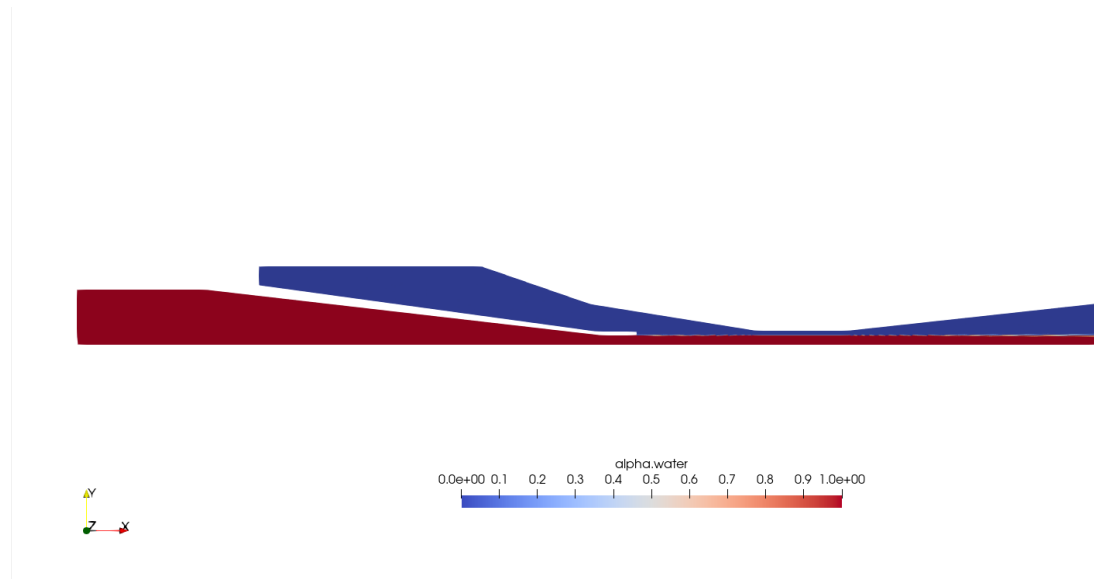


Figure 1: Water-air fraction distribution for $t = 0.2$ s

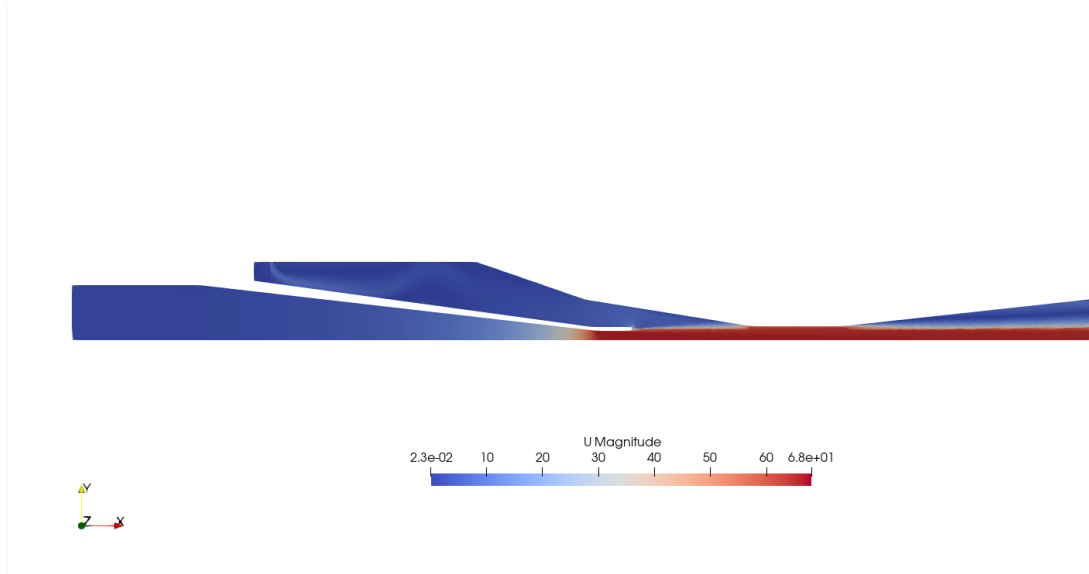


Figure 2: Velocity distribution for $t = 0.2$ s.

4 Part B: Euler-Euler Approach

The case can be also downloaded from the GitHub repository.

The step for mesh generation, setup and running of the solver are exactly the same. There are some key differences on the setup.

- Phase interaction description. In VOF there is only a configuration file with surface tension. In Euler-Euler the interaction between forces has to be completely modeled through the `constant/phaseProperties` dictionary. For a detailed description of all the items, the great Master Thesis by Otromke² can be consulted.

5 Student Exercises and Extensions

Disclaimer

This tutorial has been developed by **Robert Castilla**, of the Departament of Fluid Mechanics in the Universitat Politècnica de Catalunya, for the doctoral workshop at **IMPPAN Gdańsk** (July 2026). The author utilized **DeepSeek** for the initial definition of the document format. Further technical refinement and the development of advanced post-processing exercises were supported by **NotebookLM**. All the technical information

²Otromke, Malte. 2013. "Implementation and Comparison of Correlations for Interfacial Forces in a Gas-Liquid System within an Euler-Euler Framework."

for the **OpenFOAM v2512** environmnet and all physical interpretations were performed and approved by the author.